

## DSCI 551 Quiz 2

### Continuous distributions

- PDF  $f(x) \geq 0$ ,  $\int f(x)dx = 1$ ; density may exceed 1.  $P(X=x)=0$ ; interval probability is area:  $P(a \leq X \leq b) = \int_a^b f(x)dx = F(b) - F(a)$ .
- $F(x) = P(X \leq x) = \int_{-\infty}^x f(t)dt$ ;  $S(x) = P(X > x) = 1 - F(x)$ ; quantile  $Q(p) = F^{-1}(p)$ . CDF is nondecreasing, in  $[0, 1]$ , limits 0/1. Continuous  $\leq$  probabilities agree.
- $E[g(X)] = \int g(x)f(x)dx$ ;  $\text{Var}(X) = E[X^2] - E[X]^2$ ;  $SD = \sqrt{\text{Var}}$ . Median  $Q(.5)$ ; mode maximizes density. Central p interval:  $Q((1-p)/2)$ ,  $Q((1+p)/2)$ .
- Right-skew: usually mode < median < mean; left-skew: mean < median < mode.

### Family identification & parameters

| Family / cue                                                | Support; parameterization                                               | Mean; variance                                                             |
|-------------------------------------------------------------|-------------------------------------------------------------------------|----------------------------------------------------------------------------|
| Uniform(a,b), constant density                              | $[a, b]$ ; $f=1/(b-a)$                                                  | $(a+b)/2$ ; $(b-a)^2/12$                                                   |
| Normal( $\mu, \sigma^2$ ), symmetric bell                   | real; $\sigma$ is SD                                                    | $\mu$ ; $\sigma^2$                                                         |
| Lognormal( $\mu, \sigma^2$ ), log $X \sim N(\mu, \sigma^2)$ | $x > 0$                                                                 | $\exp(\mu + \sigma^2/2)$ ; $\exp(2\mu + \sigma^2)(\exp(\sigma^2) - 1)$     |
| Exponential(mean $\beta$ )                                  | $x \geq 0$ ; $f = \beta^{-1} \exp(-x/\beta)$ , rate $\lambda = 1/\beta$ | $\beta$ ; $\beta^2$                                                        |
| Beta( $\alpha, \beta$ ), proportion                         | $[0, 1]$ ; shapes $\alpha, \beta$                                       | $\alpha/(\alpha+\beta)$ ; $\alpha\beta/[(\alpha+\beta)^2(\alpha+\beta+1)]$ |
| Weibull(scale $\lambda$ , shape $k$ ), event time           | $x \geq 0$ ; $f = (k/\lambda)(x/\lambda)^{k-1}e^{-(x/\lambda)^k}$       | $\lambda\Gamma(1+1/k)$ ; $\lambda^2[\Gamma(1+2/k) - \Gamma^2(1+1/k)]$      |
| Gamma(shape $k$ , scale $\theta$ ), sum of waits            | $x \geq 0$ ; $f = x^{k-1}e^{-x/\theta}/(\Gamma(k)\theta^k)$             | $k\theta$ ; $k\theta^2$                                                    |

- Single next event, constant rate, memoryless  $\rightarrow$  Exponential. kth accumulated event / sum of exponentials  $\rightarrow$  Gamma. Weibull shape controls changing hazard;  $k=1$  is Exponential.  $\text{Log}(X)$  bell-shaped  $\rightarrow$  Lognormal. Proportion  $\rightarrow$  Beta; bounded constant density  $\rightarrow$  Uniform. Positive/right-skewed/waiting-time alone does not identify family.
- Do not infer parameter meaning from symbol: Exponential  $\lambda = \text{rate}$ ; Weibull  $\lambda = \text{scale}$ ; Normal  $\sigma = \text{SD}$ , not variance.

### R distribution functions

- $d*$  density,  $p*$  CDF,  $q*$  quantile,  $r*$  random draws.  $p*(q=...)$ ;  $q*(p=...)$ ; density is not point probability.
- $\text{dnorm}(x, \text{mean}=\mu, \text{sd}=\sigma)$  (pass SD);  $\text{dexp}(x, \text{rate}=\lambda)$  (rate, not mean  $\beta$ );  $\text{dweibull}(x, \text{shape}=k, \text{scale}=\lambda)$ . Other families:  $d/p/q/r + \text{unif, lnorm, beta, gamma}$ .

R call examples:  $\text{pexp}(q=3, \text{rate}=1/\beta)$  gives  $P(X \leq 3)$ ;  $1 - \text{pexp}(q=3, \text{rate}=\lambda)$  gives the upper tail.  $\text{qnorm}(.975, \text{mean}=\mu, \text{sd}=\sigma)$  returns the 97.5th percentile;  $\text{rpois}(n=1000, \text{lambda}=\lambda)$  draws 1000 counts. Always match  $p*$  input  $q$  and  $q*$  input probability  $p$ .

For a central 90% interval, use  $q*(.05)$  and  $q*(.95)$ ; for a right-tail probability  $P(X > a)$ , use  $1 - p*(a)$  or  $\text{lower.tail} = \text{FALSE}$ .

### Joint, marginal, conditional

- Joint PDF nonnegative, integrates to 1 over joint support; event probability is region integral. Marginals:  $f_X(x) = \int f_{X,Y}(x,y)dy$ ,  $f_Y(y) = \int f_{X,Y}(x,y)dx$  over legal support.
- Independence permits  $f_{X,Y} = f_X f_Y$ ; otherwise do not multiply. Draw/describe event region to set bounds.
- Event conditioning: restrict to event support,  $f_X|B(x) = f_X(x)/P(B)$  within  $B$ , zero outside. Point conditioning:  $f_Y[X(y)|x] = f_{X,Y}(x,y)/f_X(x)$ ; if independent, equals  $f_Y(y)$ . Conditional density integrates to 1 on updated support.

Example: independent  $X, Y \sim \text{Uniform}(5, 5.5)$  has joint density 4 on the support square. The event  $X < Y$  is half the square, so probability 1/2;  $P(X \leq 5.2) = 4 \times 0.2 \times 0.5 = 0.4$ . For non-rectangular events, split the integral at the support/event boundary.

Conditioning checklist: (1) restrict to the event's support; (2) divide by event probability, or for  $X=x$  divide joint density by marginal density; (3) verify the conditional density integrates to 1.

### Transformations $Y=g(X)$

- Find  $Y$  support. CDF route: rewrite  $\{Y \leq y\}$  as an  $X$  event, evaluate by original CDF/probability, split by support, differentiate.
- One-to-one:  $f_Y(y) = f_X(g^{-1}(y)) \cdot |dg^{-1}(y)/dy|$ ; multiple valid inverse branches: sum each contribution. Keep Jacobian factor.
- Example  $X \sim \text{Exp}(\text{rate}=1)$ ,  $Y = X^2$ :  $f_Y(y) = 0$  for  $y < 0$ , else  $1 - e^{-(\sqrt{y})}$ ;  $f_Y(y) = e^{-(\sqrt{y})}/(2\sqrt{y})$  for  $y > 0$ .

If  $g$  is not one-to-one, sum every valid inverse branch. Example:  $X \sim \text{Uniform}(-1, 1)$ ,  $Y = X^2$ ; for  $0 < y < 1$ , roots  $\pm\sqrt{y}$  both contribute, giving  $f_Y(y) = 1/(2\sqrt{y})$ .

### iid and likelihood / MLE

- iid sample  $y_1, \dots, y_n$ : independent, same distribution/parameters. Seed does not create iid; dependence may violate model.
- Fix observed data, vary parameter: likelihood  $L(\theta|y) = \prod_i f(y_i|\theta)$  (PMF for discrete data); it is a function of  $\theta$ , not a probability distribution over  $\theta$ .  $\ell(\theta) = \sum_i \log f(y_i|\theta)$  has same maximizer and avoids underflow.
- Grid MLE: evaluate log-likelihood on legal  $\theta$  grid; take  $\text{argmax}(\text{which.max})$ ; grid spacing gives approximation. Analytical: differentiate log-likelihood, set derivative to zero, check support/boundaries and maximum.
- Exponential(mean  $\beta$ ):  $\ell(\beta) = -n \log \beta - \sum y_i/\beta$ ;  $\hat{\beta} = \bar{y}$  (second derivative  $-n/\beta^2 < 0$ ). In R use  $\text{rate}=1/\beta$ .
- Bernoulli( $p$ ):  $\ell(p) = \sum y \log p + (n - \sum y) \log(1-p)$ ;  $\hat{p} = \bar{y}$ . For grid search keep  $p \in [0, 1]$ ; endpoints with incompatible data give  $\log(0)$ .
- Poisson( $\lambda$ ): grid concept—evaluate  $\sum \text{dpois}(y, \text{lambda}=\lambda, \text{log}=\text{TRUE})$  on legal  $\lambda \geq 0$ , maximize; distinguish observed  $y$  from candidate  $\lambda$ .

```
# Exponential mean beta: maximize the log-likelihood; beta also equals mean(y)
ll_exp <- function(beta) sum(dexp(y, rate = 1 / beta, log = TRUE))
beta_grid <- seq(0.1, 10, by = 0.01)
beta_grid[which.max(vapply(beta_grid, ll_exp, numeric(1)))]
```

```
# Bernoulli MLE grid; observed y contains 0/1 values
p_grid <- seq(0.001, 0.999, by = 0.001)
ll <- sum(y) * log(p_grid) + (length(y) - sum(y)) * log(1 - p_grid)
p_grid[which.max(ll)] # analytical estimate: mean(y)
```

### Reproducible simulation & Monte Carlo

- Define one replicate's random mechanism (including dependencies), repeat complete replicate, summarize simulated outcomes. Theoretical quantities belong to model; empirical quantities to draws ( $\text{mean}(x)$ ,  $\text{var}(x)$  with  $R$  denominator  $n-1$ ,  $\text{mean}(x==0)$ , quantiles). Finite estimates differ; LLN: iid sample mean approaches theoretical mean as  $n$  grows.
- Set seed once before experiment; same seed/RNG settings reproduce draws within a system, not necessarily across R/Python. Never reset seed inside each replicate.
- R:  $\text{set.seed}(551)$  once;  $\text{sample}(x, \text{size}, \text{replace}=\text{TRUE}, \text{prob}=\dots)$ ;  $\text{rbinom}(n=\text{draws}, \text{size}=\text{trials}, \text{prob}=p)$  ( $n=\text{draw count}$ ,  $\text{size}=\text{trials}$ );  $\text{rpois}(n, \text{lambda})$ .
- Python:  $\text{rng} = \text{np.random.default_rng}(551)$ ;  $\text{rng.choice}(x, \text{size}=n, \text{replace}=\text{True}, \text{p}=\dots)$ ;  $\text{stats.binom.rvs}(n=\text{trials}, \text{p}=p, \text{size}=\text{draws}, \text{random\_state}=\text{rng})$ ;  $\text{stats.poisson.rvs}(\text{mu}=\text{lambda}, \text{size}=n, \text{random\_state}=\text{rng})$ .
- Multi-step conditional simulation: simulate stage 1, use its result to parameterize/decide stage 2, then summarize the full replicate. E.g., attendance Bernoulli  $\rightarrow$  draw Poisson cupcakes only for attendees  $\rightarrow$  total. Do not replace random draws by conditional means or simulate outcomes for absent cases; distinguish replicate count from distribution parameters.

One-step probability estimate:  $\text{mean}(\text{simulated\_event})$  because  $\text{TRUE}=1$ ,  $\text{FALSE}=0$ ; for a Bernoulli sample,  $\text{mean}(y)$  estimates  $p$ . Empirical PMF:  $\text{table}(x) / \text{length}(x)$ . Larger independent replication usually reduces Monte Carlo variability, not to zero.

```
set.seed(551)
misses <- rbinom(n = 10000, size = 2, prob = 0.30)
p_miss_both <- mean(misses == 2) # theoretical: 0.30^2
empirical_pmf <- prop.table(table(misses))
```

Conditional mixture: if attendance  $A \sim \text{Bernoulli}(p)$  and  $C|A=1 \sim \text{Poisson}(\lambda)$ , with  $C=0$  when absent, then  $P(C=0) = 1-p+p \cdot e^{-(\lambda)}$ ; for  $c \geq 1$ ,  $P(C=c) = p \cdot e^{-(\lambda)} \lambda^c / c!$ . The mass at zero includes both absence and an attendee consuming zero.

### Solve-and-check workflow

- Identify the variable type, support, units, and data-generating process; choose PMF/PDF family from the process, not from "positive" or "count" alone.
- Probability target: discrete event  $\rightarrow$  sum PMF; continuous interval  $\rightarrow$  area or  $F(b) - F(a)$ ; right tail  $\rightarrow 1 - F(a)$ ; quantile  $\rightarrow$  solve  $F(x) = p$  / call  $q*$ .
- Bivariate event: write joint support, mark the event region, set integration bounds; factor the joint density only after independence is stated.
- MLE: observed values are fixed, parameter varies. Write one-observation PMF/PDF  $\rightarrow$  iid product  $\rightarrow$  log-likelihood  $\rightarrow$  legal parameter domain  $\rightarrow$  grid maximum or derivative/root  $\rightarrow$  verify a maximum.
- Simulation: write one full replicate in dependency order  $\rightarrow$  repeat with one seed set before the run  $\rightarrow$  summarize empirical mean/proportion/quantile  $\rightarrow$  compare to theoretical benchmark and explain finite-run variation.
- Sanity checks: probability in  $[0, 1]$ ; density integrates to 1; conditional density renormalizes on updated support; estimate lies in parameter range; distinguish replication count from distribution parameters.